**UGANDA ADVANCED CERTIFICATE OF EDUCATION**

**P510/1 PHYSICS EXAMINATIONS 2026**

**TIME: 3 HOURS**

**Instructions:**

The paper has **Six Items**. Attempt **ONLY FOUR** questions.

There are **TWO** sections; Section A and section B.

Section A is **COMPULSORY** and Section B has two parts (i) and (ii).

Attempt **NOT MORE THAN ONE ITEM** from each of the Parts (i) and (ii).

Start each Item on a fresh page.

Use the provided data and support materials to guide your solutions.

Clearly show all calculations and assumptions.

**SECTION A (COMPULSORY)**

**ITEM 1**

Netball in schools is a popular, fast-paced team sport. It emphasizes teamwork, positional discipline and quick passing, with strict rules against moving with the ball, contact, or obstruction. Janet is a senior net baller in one of the secondary schools in Uganda. Janet is of mass $79.8kg$ as shown in ***figure 1.0*** and starts her run on a rough resistive horizontal surface of the pitch developing a power of $180W$ at initial velocity of $3.6kmh^{-1}$. After running 2m, she decided to make a shot towards the goal/ring.

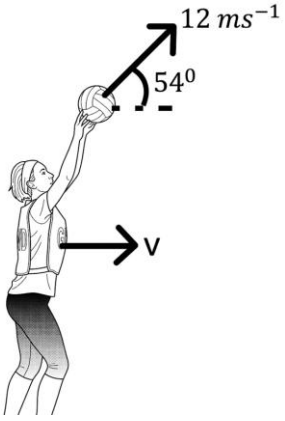



***Figure 1.0***

She throws the shot forwards with a velocity $12ms^{-1}$ with respect to her hand, in a direction $54^0$ to the horizontal at a time when her body is moving forward horizontally with a velocity $V\ ms^{-1}$.

The ball moved with the resultant velocity towards the ring while spinning. The school netball coach was amazed at her skills and he got concerned to study Janet's motion to model equations for future training drills. He decided to contact you and requested you to make him understand the origin of resistance experienced on the surface and the cause of spinning effect on the ball. He also wants to know if such a shot can enter the ring with in 5s which is 50m from the point of throw. The ring is 40cm in diameter. The school coach was also told by the Physical fitness trainer that the principle under which the ball spins has many applications

Hint:

Coefficient of friction of the surface of the pitch is 0.25.

Acceleration due to gravity is $9.81ms^{-2}$

Task:

You have been contacted for help:

(a) Address the concerns and requests made by the coach.

(b) Explain one of those applications as raised by the Physical fitness trainer.

**ITEM 2**

Uranium deposits in Uganda is currently under assessment and exploration to ensure active production for future nuclear power generation. Uganda is counting on the body of scientists to achieve this goal. One of the workers in the Ministry of atomic energy was tasked by Field Supervisor to use a radioactive detector to study the behavior of the radioactive sample and establish the type of radiation in place.

The detector has a window of effective area $250cm^2$ and is held $12.5cm$ from the rock containing a sample of radioactive $Uranium-238$. The rate meter connected to the device records $1.78x10^8Bq$. The worker was asked to: explain how the detector will do the function to determine activity and establish the possible radiations emitted and determine the mass of $Uranium-238$ in the sample.

This mass will be used to establish the amount energy produced in a nuclear plant according to the typical nuclear fission event is represented by the equation below and explain how this sample is used to produce electricity in a nuclear plant.

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{92}_{36}Kr + {}^{A}_{Z}Ba + 3{}^{1}_{0}n \quad .$$

As the worker tried moving back after data collection, he accidentally slid on one of the rocks and fell down. His arm was feared broken and his friend said he needs to be rushed to the hospital for a scan using a specific device. The worker was terrified that this device could damage him further.

Hint:

Hal-life of $\text{Uranium} - 238$ is $30$ years.

Avogadro's' constant is $6.02\text{x}10^{23}$ atoms

$$\text{Mass of } {}^{1}_{0}\text{n} = 1.67\text{x}10^{-27}\text{kg}$$

$$\text{Mass of } {}^{235}_{92}\text{U} = 390.19\text{x}10^{-27}\text{kg}$$

$$\text{Mass of } {}^{92}_{36}\text{Kr} = 152.57\text{x}10^{-27}\text{kg}$$

$$\text{Mass of } {}^{\text{A}}_{\text{Z}}\text{Ba} = 233.92\text{x}10^{-27}\text{kg}$$

Task: You have been contacted to help:

(a) provide the necessary solutions to satisfy the needs of the supervisor

(b) Assess the information given to explain how the suggested device safely works to serve the purpose.

**SECTION B (PART I)**

**ITEM 3**

Education as a fundamental human right plays a great role of empowering individuals to drive societal progress by equipping them with knowledge, critical thinking, and essential life skills. Education thus requires among others a conducive classroom environment to enhances learning for better academic performance and holistic personal growth. Available data indicates that a typical advanced level Physics student attending to the physics lesson has a heat output of $60\ W$. In Pearl High school, a physics lesson has $140$ students and lasts for $80\ minutes$. All the heat they generate is transferred to the $3200m^3$ of air in the class room. Air has a specific heat capacity of $1050\ Jkg^{-1}K^{-1}$ and average density of $1.2kgm^{-3}$. Students always complain of lack of adequate comfort and hearing during the physics lesson and they requested their class teacher to convene a class meeting with the head teacher. They requested the head teacher to install fans in the class room and increase on ventilation. The head teacher requested them to justify the need of ventilators, make him understand factors affecting sound in the room and to check if the heat generated during the lesson caused a temperature difference of at least $9^0C$ before the fans can be installed. A group of students who sit behind at the back of the class complained that for them, often do not hear properly their teacher towards the end of the lesson.

The available data indicated that their physics teacher initially produces sound represented by the equation below; $y = \text{asin}(6.6x10^3t - 20x)$ where $y$ is the displacement in meters of a wave particle at a point $x$ in meters at a time $t$. The head teacher requested the meeting to establish if the speed of this sound wave could have significantly changed by at least $50ms^{-1}$ at the end of the lesson to distort hearing. Available data indicate that class room air pressure at the end of the lesson is $1.2.0x10^5Pa$ and sound propagation in air is an adiabatic process.

Before the end of the meeting, the school's chairperson Board of governors joined. The students who sit at the back welcomed her and thereafter complained to her that they always don't see well what is written on the chalk board and requested for an optical device which uses electricity to create a large scale visual experience on the screen for everyone in class to clearly follow the lesson. The chairperson asked for more information about the device.

HINT:

The ratio of the principal specific heats of the gas $\gamma = 1.4$

Task

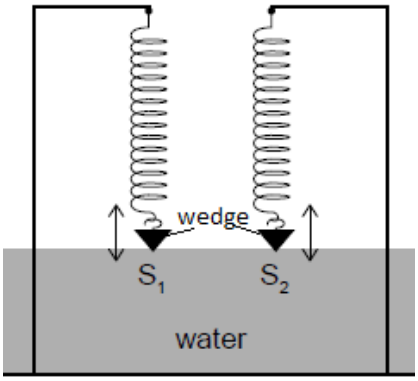
As a student of physics;

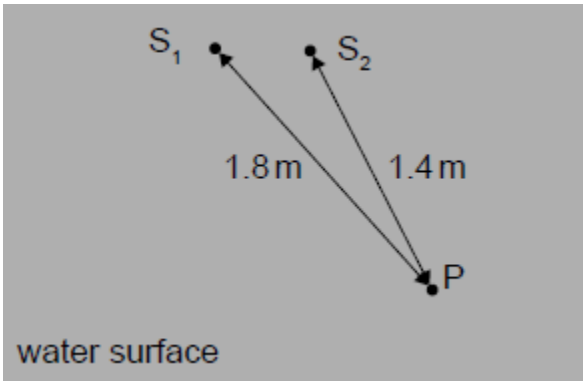
(a) Interpret the information given to address the requests made by the head teacher.

(b) Assess the request of the Board Chair person by describing the working of the suggested device.

**ITEM 4**

Energy is a fundamental property of nature that cannot be created or destroyed, only transformed from one form to another for various life applications. A group of S.6 Physics students decided to perform some experiments to enable them understand concepts on energy better. Okello, one of the group members used two identical wedges attached to two identical vertical springs. The wedges touch a water surface at S1 and S2. As the wedges oscillate, they create travelling waves on the water surface. The wedges oscillate in phase with the same amplitude. P is a point on the surface of the water.



The mass of each wedge is $0.200\ kg$ and the spring constant is $504\ N\ m^{-1}$. Damping is negligible. The speed of the water waves is $1.6\ m\ s^{-1}$. Okello was tasked by the group to find the frequency of the water wave and make them understand the observations made at point P and slightly after.

The head teacher also had a newly purchased a $2.352kW$ power Kettle made of a material of heat capacity of $420\ JK^{-1}$ and he was assured by the shop owner that the kettle was so genuine and can boil water within $6\ minutes$. The head teacher gave it to the group to test whether it was truly genuine. They heated $2\ litres$ of water initially at $20^0C$ until it boiled at $100^0$. Okello observed that the steam produced had energy and was almost lifting off the cover. The discussion leader asked the group to heat the water for extra $4\ minutes$ and establish the work done by the water against a constant atmospheric pressure of $1.0x\ 10^5Pa$. The head teacher came back and asked them to give the molecular account of all the heating processes.

Rose entered the room and found a great science group. She requested them to recommend a device which can help her vision. She was always complaining of her failure to see far writings clearly.

HINT

Density of water is $1000kgm^{-3}$ while that of steam is $0.6272kgm^{-3}$.

Specific heat capacity of water is $4200Jkg^{-1}K^{-1}$

Specific latent heat of vaporization of steam is $2.25x10^6Jkg^{-1}$

As a student of physics;

(a) Analyze the scenario to address the concerns raised in the discussion group.

(b) Recommend the device for Rose and make her understand how it will be of help.

## SECTION B (PART II)

### ITEM 5

Some schools in Uganda are faced with challenges of ineffective communication especially on assemblies due to lack of address systems. The head teacher in one of the Secondary Schools bought a portable speaker address system. This speaker could however, work for a short time and stop. The head teacher claimed that the problem of the speaker could be a high internal resistance of the battery used for charging. Some learners claimed that it was the charging time which was short. The recommended charging time for the power bank to be full should be at least 15 minutes. One S.5 student approached the head teacher and

requested to test the battery and establish if it was indeed the problem. The student connected the battery in series with a variable resistor and an ammeter. When the resistor was set $21\Omega$, the current through it was $0.48A$ and when it was set to $36\Omega$ the current was $0.30A$.

The power bank in the speaker was made up of two super capacitors each of capacitance $1.0F$ in series. The head teacher further intervened and asked how the capacitors got charged by the battery and the time one needed to wait before the speaker is fully charged.

The Deputy head teacher on the other hand asked the student to explain how he could change the a.c from other sources into d.c for use in powering some other school gadgets.

The S.5 student tried but could not succeed in addressing any of the matters raised. He went to S.6 physics students for assistance.

**Hint** A good battery for charging has internal resistance not more than $20\Omega$.

Task:

You have been selected to help:

(a) analyze the information given in the scenario and address the claims and concerns of the head teacher.

(b) Address the request of the deputy head teacher.

**ITEM 6**

Listening to religious radios in highly advisable for older adults to promote emotional comfort and mental stimulation. Opio visited his parent with his child who was doing physics in S.6. The parent complained to Opio of the challenge they had with their radio set which often failed to tune in their favorable religious program. The radio operated at a frequency of $61.2MHz$. Opio opened the radio and found its tuning circuit as indicated in figure 6(i) very weak.

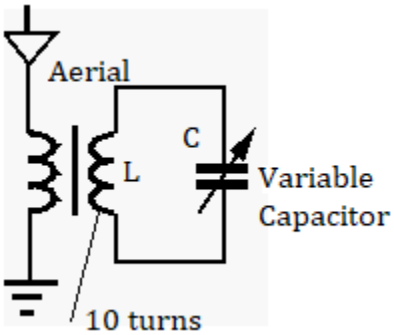



Figure6 (i): Resonance circuit for tuning radio.

Opio decided to install a new stronger variable capacitor, C and a pure inductor,L. He purchased materials and made the two smaller capacitors each of plate separation $2\text{cm}$ and maximum area of overlap

$5.0\text{x}10^{-4}\text{m}^2$ filled with a material of dielectric constant 4. The two capacitors were joined in series and connected with a rotor for tuning.

He made L from a $1.0\text{cm}$ air-cored solenoid of $10\ \text{turns}$ and $1.2\text{x}10^{-3}\text{m}^2$ cross sectional area. Opio requested his child to explain the effect of a dielectric in the capacitors, determine the magnetic field density developed by L and perform necessary calculations to establish whether the new circuit can tune to the desired religious channel

Opio's Parent also took advantage of the presence of the grand child and requested to make them understand the structure and operation of a device used by used in charging the radio because, this parent had always wondered how the charger can be connected in the socket which supplies $240\text{V}$ to charge a radio which operates at only $12\text{V}$.

Hint: Permeability in free space $\mu_0 = 4\pi\text{x}10^{-7}\text{Hm}^{-1}$

Permittivity of free space $\varepsilon_0 = 8.85\text{x}10^{-12}\text{Fm}^{-1}$

Task: As a S.6 physics student:

(a) analyze the information in the scenario to answer the requests made by Opio

(b) Provide the solution made by Opio's parent.

# P510/2

**ITEM FROM FORCES AND MOTION (MECHANICS)**

A firm and strong roof of a large Warehouse requires long nails to fix it but the weight of the nails must be cost effective since this weight does not affect the strength of the roof. The chief Engineer of a construction Company contracted to construct a factory infrastructure instructed the workers fixing the roof that the nails to be used must each have an average mass in the range 25.0 g to 60.0 g for the Company to realize profits. One of the Company Foremen bought the nails for roofing phase but the shopkeeper and the Foreman did not know how to verify the average mass of each nail. The Engineer has contacted you and given you a sample nail. He wants you to help the Company establish whether the nails can be used for roofing the factory. He has requested you to use two methods. One of the methods requires determining mass of one nail using a known mass value of 0.250 kg and a metre rule. The other method requires use of varying masses and data analysis is done using a graph to obtain another value of mass of the nail.

**Task:**

Design and carry out an investigation using the two methods to establish if the nails can be used for roofing the factory.

**Hint:** The mass $m$ of each nail is related to period $T$ of oscillation by the expression:

$$(M + m)\pi^2 = \frac{T^2K}{4}$$

where $K$ is the spring constant.

**ITEM FROM ENERGY (LIGHT)**

Opticians play a vital role of finding solutions for people who have sight problems.
In the neighboring clinic, a Nurse got a sight defect where she could only read patients' prescriptions by extending the writing far away from her eyes. She went to the optician for help. The optician recommended that she should use spectacles containing convex lenses of mean power in the range $+\mathbf{12.5D}$ to $+\mathbf{7.1D}$ to be able to read the writings when they are close to her eyes. The following day, the Nurse could see the writing while using the spectacles but did not have enough clarity. She went back to the optician with a claim that the power of the lenses in the spectacles did not lie in the recommended range. The optician did not have the apparatus to establish the power of the lenses. He has given you a sample lens from the spectacles to help and find out if the Nurse's claim is correct. For accuracy, you have been told to use two methods.
The first method involves a plane mirror and a screen with a wire gauze in a hole drilled at its center, used to obtain the lens power.
The second method involves use of varying separation of two positions of the lens that can result into formation of an image on a white screen for different separations of the object and the screen. Graphical analysis is also done in this method to obtain another value of power of the lens.
Task:
Carry out a scientific investigation using the two methods to establish whether the Nurse can continue to use the spectacles.
Hint:
The object distance, $u$ from the lens is related to the separation, $x$ of the two positions of the lens and separation, $y$ of the object and screen by the expression:

$$U = \frac{(y - x)}{2}$$

**ITEM FROM FIELDS AND CHARGES (CURRENT ELECTRICITY)**

Ironed clothes on a ceremony is part of what makes a person look smart and admirable. Students wanted to iron their clothes as part of preparations for their Leavers' party but the flat iron failed to work. They took the flat iron to the school laboratory technician for repair. The technician opened the flat iron and found the wire that provides heat had been broken. He sent one of the students to buy a thin bare wire of resistivity in the range $4.6 \times 10^{-7}\Omega\ \mathrm{m}$ to $1.3 \times 10^{-6}\Omega\ \mathrm{m}$ to fix in the flat iron as a replacement for the broken wire. The student bought the wire, unfortunately it did not have resistivity value indicated on it. The laboratory technician is arranging some apparatus for a practical lesson. He has requested you for help to determine the resistivity of the purchased wire to find out whether it is suitable for use as a replacement. He has also proposed two methods to be used, one method requires you to use a single known length equal to 0.600 m of the bare wire connected to an ammeter and voltmeter. In the other method, varying lengths of the thin bare wire is connected in one gap of the metre bridge and a standard resistor is connected in the other gap, and results are analyzed using a graph. The two values of resistivity are averaged.

Task:
Plan and conduct an investigation using the two methods to verify the suitability of the bare wire purchased.

**End.**